

Utilizing Very High-resolution Optical RGB Satellite Imagery in Geo-information Extraction for Fine-scale Map-making

Wenmiao Hu

Grab-NUS AI Lab, National University of Singapore
Singapore
hu.wenmiao@u.nus.edu

ABSTRACT

Maps are the fundamental elements of any navigation and localization system. With the fast expansion of urban areas and the increasing complexity of modern cities, traditional mapping techniques cannot meet the need for frequent map updates with enriched map details. This work investigates how geo-referenced very high-resolution (VHR) RGB satellite imagery can be used to extract geo-information to support the creation of fine-scale maps and facilitate map updates, focusing on two sub-directions. First, to alleviate the data scarcity issue of large-scale geo-information extraction from satellite imagery-based datasets, (1) GAN-assisted road segmentation proposes a new assisted training scheme to improve the model performance when the training dataset is limited and (2) a context-enhanced satellite-imagery dataset is created for large-scale parking lot detection to improve the type diversity of target objects. Second, to support rich map attribute geo-information extraction and vision-based navigation using street-view imagery, new methods are proposed to improve location and orientation extraction of the street-view imagery via cross-view matching with satellite imagery.

CCS CONCEPTS

• **Information systems** → **Information extraction; Image search;**
• **Computing methodologies** → **Image segmentation; Transfer learning; Scene understanding.**

KEYWORDS

GAN; Data augmentation; Image segmentation; Satellite imagery; Street-view imagery; Cross-view matching; Geo-localization; Geo-information extraction.

ACM Reference Format:

Wenmiao Hu. 2024. Utilizing Very High-resolution Optical RGB Satellite Imagery in Geo-information Extraction for Fine-scale Map-making. In *Proceedings of the 32nd ACM International Conference on Multimedia (MM '24)*, October 28–November 1, 2024, Melbourne, VIC, Australia. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3664647.3685647>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

MM '24, October 28–November 1, 2024, Melbourne, VIC, Australia

© 2024 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0686-8/24/10

<https://doi.org/10.1145/3664647.3685647>

1 INTRODUCTION

Maps are the fundamental element of any navigation and localization system [19, 20, 32]. They also play critical roles in other fields such as urban planning and city re-development [25, 26, 28]. To support the increasing demand for location-based services and abundant map-based applications that require fine-grained map information, this work focuses on geo-information extraction for fine-scale map-making. A fine-scaled map shall contain detailed information on every significant object in a large area needed for applications like navigation and urban planning. For example, a map contains boundaries (*geometrical* info) of key objects at meter-level granularity, connectives (*topological* info) among different objects and rich information (*attributes*) such as shop names for buildings and turning restrictions for roads.

To combine fine granularity with large ground coverage, traditional mapping takes a great amount of manpower and time to complete one map. It cannot meet the need for frequent map updating, especially with the fast expansion of urban areas and the increasing complexity of modern cities. Fortunately, benefiting from the advancement of remote sensing, IOT systems, map digitization and affordable consumer-grade devices, multiple data sources can be used to extract geo-information. Among them, satellite remote sensing observes our Earth from space. Compared to images collected by ground devices or airborne data (e.g., from Unmanned Aerial Vehicles), satellite imagery can be collected regularly with no additional local access permission and low human interaction once the satellites are deployed into the orbits. The bird-view images have large ground coverage, which is ideal for large-scale surveys. They present the optimal source to provide consistent, frequent, large-scale assessments [29].

In this work, we investigate the possibility of using VHR (Very High Resolution) optical RGB satellite images to extract useful geo-information for map-making. Although satellite remote sensing gives different options for data types, we focus this study on the usage of VHR, below 1 meter/pixel GSD (Ground Sampling Distance), *optical* RGB satellite imagery, which 1) gives the best fit to the dimension of the target objects for fine-scale map-making; 2) is closest to human perceptions for possible human-interactive and multi-modality applications and validations; 3) is one of the most common types of imagery collected, pre-processed and archived for historical and future purchases in the commercial market. In the remainder of the work, it is referred to as “satellite image” or “satellite imagery”.

Although there are previous works on geo-information extraction from satellite imagery, we find the three main challenges to support fine-scaled map-making within the context of multimodal data availability:

- For satellite imagery-based information extraction, the existing datasets are limited in terms of quantity and diversity. We need to provide more datasets and try to alleviate the data scarcity issue with alternative solutions.
- There are different data from different sensors that contain useful information for map-making. However, due to temporal, spatial, and format discrepancies, those data cannot be merged easily. We need to find efficient ways to fuse such data to optimize the cohesion of geo-referenced data.
- With the reduced cost and size of portable devices, crowd-sourced data, especially street-view imagery, can be easily collected. They contain rich and detailed information for map-making, which cannot be replaced by other data sources. However, those images usually come with no/missing location and orientation information. We want to retrieve/improve the missing/noisy information so that these images can be used for location-based services or map-making.

With these considerations, this work proposed different methods and a new dataset to demonstrate the usage of satellite imagery solely and/or combined with other data sources. The main contributions can be categorized into two sub-directions:

- Provide new methods and a parking-lot detection dataset to alleviate the data scarcity issues to support better performance on large-scale surveys on geometrical and topological geo-information extraction.
- Propose new methods to improve the granularity of the orientation extraction of street-view imagery query along with geo-localization to support local-scale attribute estimation and visual navigation.

The rest of this paper presents the contributions that have been achieved for the targeted objectives, the resources available to the research communities and the outlook of this domain. The full contents of the sub-works can be found in the publications¹ GAN-Assisted Training [11], GeoPalette [10], Grab-Pklot dataset [31], Beyond Geo-Localization [13] and PetalView [12].

2 APPROACHES AND RESULTS

2.1 Satellite Imagery-based Geo-information Extraction with Data Scarcity Issues

In recent years, information extraction from satellite imagery has attracted increasing attention, and new datasets have been released for different use cases [7, 24, 25, 33]. For supervised learning, a dataset minimally contains both the input data and the outcome ground-truth. However, there are limited high-resolution satellite datasets available because of the cost of satellite imagery acquisition and annotation. Compared to close-range imagery datasets such as ImageNet [8], existing satellite datasets have a much lower number of images and cover only a few scenarios (cities, background environments, and target objects). To alleviate the data scarcity issue, GAN-assisted training and Grab-Pklot dataset are proposed.

2.1.1 GAN-assisted Training. This project aims to use synthetically generated samples to improve the model performance when the

¹Appreciate the contribution of my co-authors Y. Yin, Y. Zhang, Y. Liang, X. Han, A. Georgescu, Y. K. Tan, A. Tran, S.-K. Ng, H. Kruppa, and R. Zimmermann in these publications.

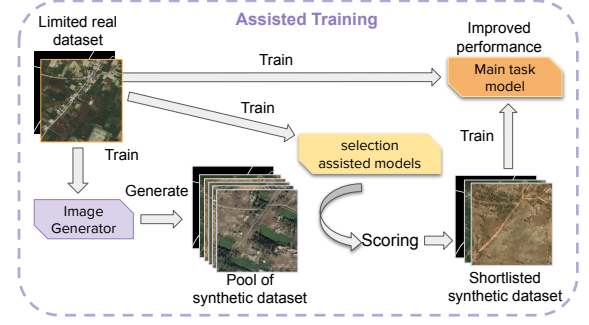


Figure 1: Illustration of GAN-assisted training when the real training dataset is limited. (Real sample from DeepGlobe [7]).

Table 1: Synthetic data assisted trained segmentation models performance compared to baselines.

Generator	Test mIoU	To B1	Test mIoU	To B1
Baseline B1	60.920%	-	-	-
Baseline B2	64.590%	3.670%	64.590%	3.670%
Strategy	Self-augmentation		Scene-creation	
Real:Fake Ratio 1:3 R3				
cSinGAN	64.440%	3.520%	64.420%	3.500%
GeoPalette	64.320%	3.400%	64.160%	3.240%
Real:Fake Ratio 1:1 R1				
cSinGAN	64.090%	3.170%	64.220%	3.300%
GeoPalette	63.940%	3.020%	64.380%	3.460%

real dataset is limited. The proposed GAN-assisted training scheme (Figure 1) has three key components : synthetic image generation, sample selection and assisted training. In this project, new image generators and a set of new synthetic sample evaluation metrics are proposed along with extensive experiments on different configurations of image generating and mix training strategies. The proposed image generators, cSinGAN and Geo-Palette, can augment existing images at the appearance level (self-augmentation) and create new images with novel scene structures that are not covered by the real dataset (scene-creation). Our results on road segmentation (Table 1) show that indeed using synthetic training sample pairs can improve overall performance when the real training dataset is limited. Experiments are conducted on DeepGlobe [7]. The baseline B1 and B2 are standard trained with 1000 and 4000 real samples, and there is a 3.67% performance gap caused by the reduced number of real samples used in the training. After using synthetically generated samples generated from our cSinGAN and GeoPalette, the assisted trained models almost reached B2, achieving a 4 times reduction in the number of real samples used in the training. More details can be found in GAN-Assisted Training [11] and GeoPalette [10].

2.1.2 Grab-Pklot Dataset. The second effort to extend the coverage of the existing datasets is the creation of the Grab-Pklot Dataset. Grab-Pklot dataset is the first high-resolution and context-enriched satellite imagery dataset for large-scale parking lot detection. Compared to related work [1, 2, 6, 14], Grab-Pklot contains high-resolution satellite imagery from bird view and the corresponding annotation at the parking lot level, dedicated to large-scale survey without prior knowledge of the location of parking lots;

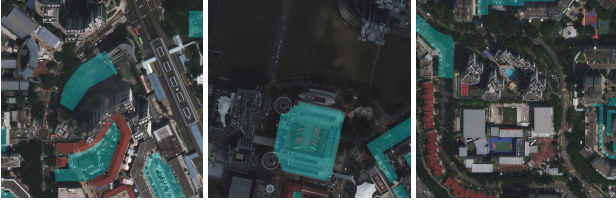


Figure 2: Grab-Pklot samples from train (left), test (middle) and additional data (right) splits. Original Images © 2018 Maxar Technologies Inc.

on top of the image and annotation, the neighbourhood roads and buildings information are given as contextual input for multimodal data fusion study.

The dataset contains a total of 1,344 satellite images in Singapore. The satellite samples have a size of 1024×1024 pixels with a ground sampling distance of 0.3 meter/pixel. Samples are shown in Figure 2. As the parking lot usually has irregular shapes and can be occluded by vegetation, it is sometimes difficult to determine the exact boundary for some of the parking lot instances. Parking lots with heavy vegetation occlusion or roof cover are not labeled, as well as single-line sidewalk parking. The dataset is split into a training set of 527 samples, a testing set of 200 samples, and an additional training set of 617 samples. The images in the training and testing sets are labeled with manually refined high-quality parking lot annotations. The images in the additional training set are labeled with partial parking lot annotations where missing parking blocks/lots exist. The additional training set reflects the quality of parking lot annotations in real-world crowdsourced maps. This work also gives a baseline method to fuse the contextual information with the satellite imagery input. We found using contextual information generally improves the performance and observed a mIoU improvement of 1.80% ~ 3.18% by our proposed fusion-based segmentation approach on various backbone methods [3–5, 15, 17, 21, 35, 36]. More details can be found in Grab-Pklot dataset [31].

2.2 Street-view Imagery Geo-localization via Cross-view Matching with Satellite Imagery

The second direction in this work is to extract the location and orientation information of a given street-view query via cross-view matching with satellite imagery. At the beginning of this study, most of the existing work at the moment focused on location estimation. However, for map-making purposes, both location and orientation extraction are equally important. Hence in this stream of work, new methods are proposed to improve the orientation estimation of street-view imagery along with geo-localization in large-scale search (image-retrieval-based search from a pool of satellite imagery candidates) and local-scale search (fine location extraction within one satellite imagery when location prior is given).

2.2.1 Beyond Geo-localization: fine-grained orientation extraction in large-scale search. This project aims to improve the granularity of the street-view image orientation extraction in large-scale search. To cater to the fine-grained orientation extraction problem, the following contributions are introduced: 1) two new reference coordinates for orientation misalignment and absolute estimated

Table 2: Orientation estimation on two datasets for large-scale search. The baseline DSM [23]* is used to compute the gain in the brackets. $r@2^\circ$ means the percentage of images in the dataset that have an estimated error below 2 degrees.

Model	Mean error ↓	r@2° (%)	r@5° (%)	Mean error ↓	r@2° (%)	r@5° (%)
	CVUSA			CVACT		
Zhai [34]	-	≤15	-	-	-	-
Zhu [37]	-	≤24	-	-	-	-
DSM [23]*	5.29°	47.49	93.25	6.26°	44.13	88.31
Ours (FI) [13]	3.78°	82.13	96.77	4.88°	70.87	92.29
Gain	(1.51°)	(34.64)	(3.52)	(1.38°)	(26.74)	(3.98)
Ours (CS) [13]	3.77°	82.42	96.75	4.75°	72.28	92.43
Gain	(1.52°)	(34.93)	(3.50)	(1.51°)	(28.15)	(4.12)

error, 2) new evaluation metrics, 3) two methods to improve the granularity of the orientation extraction without introducing new trainable parameters and 4) a new absolute angle loss.

Our proposed methods Curve Smoothing (CS) and Feature Interpolation (FI) can obtain sub-pixel orientation estimation on the orientation curve [23] with a flexible scaling number that can be adapted to different use cases. The proposed angle loss gives direct supervision on the orientation estimation. Experiments are conducted on two datasets: CVUSA [27] and CVACT [18]. Compared to previous methods, both CS and FI achieve significantly better performance on the orientation extraction (Table 2). Additionally, by integrating the fine-grained orientation extraction, the performance of geo-localization also increased. Compared to DSM [23]*, $r@1$ (recall at top 1) image retrieval test with unknown orientation on CVUSA improved by 4.70% for FI and 4.64% for CS; on CVACT are improved by 5.17% for FI and 5.55% for CS. More details can be found in Beyond Geo-localization [13].

2.2.2 PetalView: fine-grained meta information extraction in local-scale search. As a follow-up work, PetalView refines the problem to local-scale search within one satellite imagery and aims to obtain fine-grained orientation and location information of street-view imagery query at sub-degree and sub-meter level. The reasons for investigating fine-grained orientation and location prediction in *local-scale search* within one reference satellite image are twofold: 1) it is a standalone solution to refine the location and orientation information when a prior location in other modalities is given (e.g., GPS, text tags), and 2) it can be used as a complementary step following a large-scale search. Compared to previous and concurrent work [16, 22, 30], PetalView aims to provide fine-grained location and orientation extraction around a close neighbourhood around a location prior with or without an orientation prior to guide the orientation extraction; the new method disentangles the feature extraction process for the two views which potentially can be applied for large-scale search.

Figure 3 shows the conceptual illustration of the PetalView representation and multi-scale search strategy. To boost the finer-grained search, we propose to extract the PetalView feature, comprising multiple petals, which are defined based on viewing angles. Furthermore, each petal contains information dissected into different zones by distance. Each anchor in the satellite view has limited access to pixels of the image feature map based on the relative distance and angle direction. Paired with the multi-scale search strategy, which

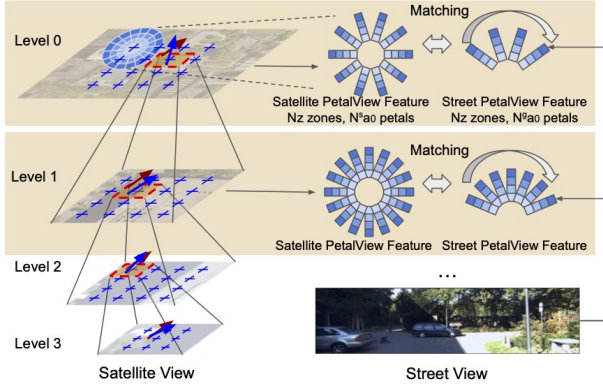


Figure 3: PetalView with multi-scale search in local-scale search. Blue arrow: predicted direction & location. Brown arrow: ground-truth. Blue cross: search anchors.

Table 3: Performance on VIGOR dataset with no angle prior in local-scale search.

	Same Area		Cross Area	
	Angle ($^{\circ}$) $\epsilon_{\text{Mean}} \downarrow$	Location (m) $\epsilon_{\text{Mean}} \downarrow$	Angle ($^{\circ}$) $\epsilon_{\text{Mean}} \downarrow$	Location (m) $\epsilon_{\text{Mean}} \downarrow$
MCC [30]	56.86	9.87	72.13	12.66
SliceMatch	25.46	6.49	25.97	7.22
ResNet50 [16]				
Ours [12]	19.15	4.58	24.09	5.91

uses the best-matched anchor as the search center for the next level search and dynamically deploys finer grids, we enable pixel-level search precision in the feature space without an exhaustive search at every pixel in the extracted feature maps. Our method achieves higher and more stable overall performance through different use cases. The proposed method is tested on two datasets: the selected KITTI-Satellite dataset [9, 22] and the VIGOR dataset [16, 38]. Table 3 shows the results on VIGOR tests². For all reported metrics in the previous paper, we obtain the best performance, achieving mean error on orientation and location $19.15^{\circ}/24.09^{\circ}$ and $4.58/5.91$ meters for the two tests. Figure 4 shows an illustration of multi-scale search with the results. Given a search area of $[128, 128]$ in the feature space, using a four-level search reduces the number of queries from $16,384 = 128^2$ times to $57 = 3 \times 16 + 9$ times and the accuracy increases drastically at each level. More details can be found in PetaView [12].

3 RESOURCES AND OUTLOOKS

For academic non-profit research purposes, the codebase and dataset can be released based on request after a usage review. Currently, the Grab-Pklot and the codebase of Beyond-Geo-Localization are ready for request. The instructions on resource applications can be found on the GitHub page of the author³.

Geo-information-related services have already played a very important role in our daily lives. With the development of new technologies, such as large language models, autonomous devices,

²We reprint the results of MCC [30] and SliceMatch [16] reported in SliceMatch [16].

³Note that, to comply with Grab and third-party requirements, the applicants need to explicitly state the usage and sign the corresponding non-disclosure agreement and/or academic research license before obtaining the resources.

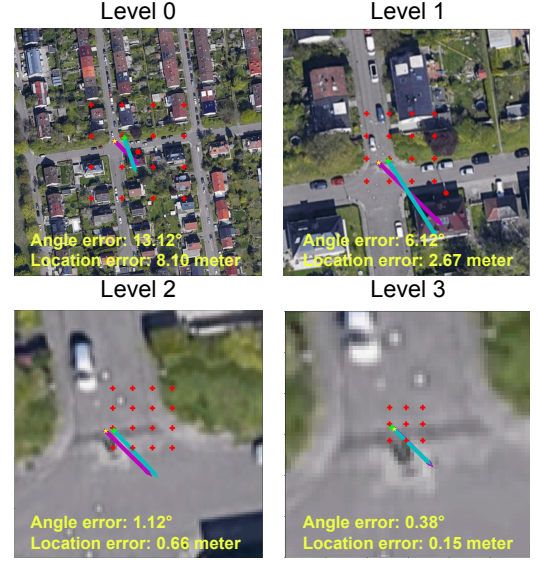


Figure 4: Anchors at each level in multi-scale search with prediction results. Cyan arrow: detected direction & location. Pink arrow: ground-truth. Red cross: anchors. Red circle: image center. (Original images from KITTI-Satellite dataset [9, 22])

virtual reality and mixed realities, we look forward to new use cases, interactive systems and higher performance of the new generation of applications in the geospatial domain. In general, there are the following aspects we shall look into:

- Performance improvement and efficient training with limited datasets. Improve the robustness and reduce time cost to support real-time process systems that handle data captured in the wild across different geographic regions.
- Machine-assisted tools that can improve productivity and automate tasks where possible for map-making and dataset creation.
- Multi-source multi-timestamp confident-weighted data fusion with 5D representation (3D spatial + time + attribute) to support regular map updates.
- Rich map that contains information that could be beneficial for machine users but beyond the user cases for human users.

4 CONCLUSION

This work explores the usage of VHR satellite imagery in geo-information extraction for fine-scale map-making from two different aspects. Firstly, satellite images can be used as the primary data source to extract the geometrical and topological information of map target objects. Then, to extract the missing attribute information from street-view imagery, satellite imagery can be used for geo-localizing the street-view imagery before the fine-grained information extraction. The outcomes of this work merge and contribute to the overall goal of creating and updating fine-scale maps that can be used to enhance existing location-based services, providing updated information for decision-making and future planning.

ACKNOWLEDGMENTS

This work was funded by the Grab-NUS AI Lab, a joint collaboration between GrabTaxi Holdings Pte. Ltd. and National University of Singapore, and the Industrial Postgraduate Program (Grant: S18-1198-IPP-II) funded by the Economic Development Board of Singapore. I would like to thank Maxar Technologies Inc. for allowing the original satellite images to be released as part of Grab-Pklot dataset for non-profit academic research purposes. I appreciate the contribution of my co-authors Yifang Yin, Yichen Zhang, Yuxuan Liang, Xianjing Han, Andrei Georgescu, Ying Kiat Tan, An Tran, See-Kiong Ng, Hannes Kruppa, and Roger Zimmermann in the related publications. I also need to express my greatest gratitude to my Ph.D. supervisor Prof. Roger Zimmermann for his guidance and support in my study and the support from my colleagues Aashish Nikhil Ghosh, Jinming Cao, Na Zheng.

REFERENCES

- [1] Giuseppe Amato, Fabio Carrara, Fabrizio Falchi, Claudio Gennaro, Carlo Meghini, and Claudio Vairo. 2017. Deep learning for decentralized parking lot occupancy detection. *Expert Systems with Applications* 72 (2017), 327–334.
- [2] Giuseppe Amato, Fabio Carrara, Fabrizio Falchi, Claudio Gennaro, and Claudio Vairo. 2016. Car parking occupancy detection using smart camera networks and Deep Learning. In *2016 IEEE Symposium on Computers and Communication (ISCC)*. 1212–1217.
- [3] Abhishek Chaurasia and Eugenio Culurciello. 2017. LinkNet: Exploiting encoder representations for efficient semantic segmentation. In *2017 IEEE Visual Communications and Image Processing (VCIP)*. 1–4.
- [4] Liang-Chieh Chen, George Papandreou, Iasonas Kokkinos, Kevin Murphy, and Alan L. Yuille. 2018. DeepLab: Semantic Image Segmentation with Deep Convolutional Nets, Atrous Convolution, and Fully Connected CRFs. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 40, 4 (2018), 834–848.
- [5] Liang-Chieh Chen, Yukun Zhu, George Papandreou, Florian Schroff, and Hartwig Adam. 2018. Encoder-Decoder with Atrous Separable Convolution for Semantic Image Segmentation. In *Computer Vision – ECCV 2018*, Vittorio Ferrari, Martial Hebert, Cristian Sminchisescu, and Yair Weiss (Eds.). Springer International Publishing, Cham, 833–851.
- [6] Paulo R.L. de Almeida, Luiz S. Oliveira, Alceu S. Britto, Eunelson J. Silva, and Alessandro L. Koerich. 2015. PKLot – A robust dataset for parking lot classification. *Expert Systems with Applications* 42, 11 (2015), 4937–4949.
- [7] Ilke Demir, Krzysztof Koperski, David Lindenbaum, Guan Pang, Jing Huang, Saikat Basu, Forest Hughes, Devis Tuia, and Ramesh Raskar. 2018. DeepGlobe 2018: A Challenge to Parse the Earth through Satellite Images. In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*. 172–17209.
- [8] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. 2009. ImageNet: A large-scale hierarchical image database. In *2009 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 248–255.
- [9] A Geiger, P Lenz, C Stiller, and R Urtasun. 2013. Vision meets robotics: The KITTI dataset. *The International Journal of Robotics Research* 32, 11 (2013), 1231–1237.
- [10] Wenmiao Hu, Yifang Yin, Ying Kiat Tan, An Tran, Hannes Kruppa, and Roger Zimmermann. 2021. GeoPalette: Road Segmentation with Limited Satellite Imagery. In *Proceedings of the 29th International Conference on Advances in Geographic Information Systems (Beijing, China) (SIGSPATIAL '21)*. Association for Computing Machinery, New York, NY, USA, 95–98.
- [11] Wenmiao Hu, Yifang Yin, Ying Kiat Tan, An Tran, Hannes Kruppa, and Roger Zimmermann. 2023. GAN-Assisted Road Segmentation from Satellite Imagery. *ACM Trans. Multimedia Comput. Commun. Appl.* (Nov. 2023).
- [12] Wenmiao Hu, Yichen Zhang, Yuxuan Liang, Xianjing Han, Yifang Yin, Hannes Kruppa, See-Kiong Ng, and Roger Zimmermann. 2023. PetalView: Fine-Grained Location and Orientation Extraction of Street-View Images via Cross-View Local Search. In *Proceedings of the 31st ACM International Conference on Multimedia*. Association for Computing Machinery, New York, NY, USA, 56–66.
- [13] Wenmiao Hu, Yichen Zhang, Yuxuan Liang, Yifang Yin, Andrei Georgescu, An Tran, Hannes Kruppa, See-Kiong Ng, and Roger Zimmermann. 2022. Beyond Geo-Localization: Fine-Grained Orientation of Street-View Images by Cross-View Matching with Satellite Imagery. In *Proceedings of the 30th ACM International Conference on Multimedia*. Association for Computing Machinery, New York, NY, USA, 6155–6164.
- [14] Nisim Hurst-Tarrab, Leonardo Chang, Miguel Gonzalez-Mendoza, and Neil Hernandez-Gress. 2020. Robust Parking Block Segmentation from a Surveillance Camera Perspective. *Applied Sciences* 10, 15 (2020).
- [15] Alexander Kirillov, Kaiming He, Ross Girshick, and Piotr Dollár. 2017. A unified architecture for instance and semantic segmentation. In *CVPR*.
- [16] Ted Lentsch, Zimin Xia, Holger Caesar, and Julian F. P. Kooij. 2023. SliceMatch: Geometry-Guided Aggregation for Cross-View Pose Estimation. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 17225–17234.
- [17] Hanchao Li, Pengfei Xiong, Jie An, and Lingxue Wang. 2018. Pyramid Attention Network for Semantic Segmentation. In *British Machine Vision Conference 2018, BMVC 2018, Newcastle, UK, September 3–6, 2018*. BMVA Press, 285.
- [18] Liu Liu and Hongdong Li. 2019. Lending Orientation to Neural Networks for Cross-View Geo-Localization. In *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 5617–5626. <https://doi.org/10.1109/CVPR.2019.00577>
- [19] Mohammed A. Quddus, Washington Y. Ochieng, and Robert B. Noland. 2007. Current map-matching algorithms for transport applications: State-of-the-art and future research directions. *Transportation Research Part C: Emerging Technologies* 15, 5 (2007), 312–328.
- [20] Krishna Regmi and Mubarak Shah. 2021. Video Geo-Localization Employing Geo-Temporal Feature Learning and GPS Trajectory Smoothing. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*. 12106–12115.
- [21] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. 2015. U-Net: Convolutional Networks for Biomedical Image Segmentation. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, Nassir Navab, Joachim Hornegger, William M. Wells, and Alejandro F. Frangi (Eds.). Springer International Publishing, Cham, 234–241.
- [22] Yujiao Shi and Hongdong Li. 2022. Beyond Cross-view Image Retrieval: Highly Accurate Vehicle Localization Using Satellite Image. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 16989–16999.
- [23] Yujiao Shi, Xin Yu, Dylan Campbell, and Hongdong Li. 2020. Where Am I Looking At? Joint Location and Orientation Estimation by Cross-View Matching. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 4063–4071.
- [24] Adam Van Etten, Dave Lindenbaum, and Todd M. Bacastow. 2018. SpaceNet: A Remote Sensing Dataset and Challenge Series. *CoRR* abs/1807.01232 (2018).
- [25] Junjie Wang, Zhuo Zheng, Ailong Ma, Xiaoyan Lu, and Yanfei Zhong. 2021. LoveDA: A Remote Sensing Land-Cover Dataset for Domain Adaptive Semantic Segmentation. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, J. Vanschoren and S. Yeung (Eds.), Vol. 1. Curran.
- [26] Mingshu Wang and Floris Vermeulen. 2021. Life between buildings from a street view image: What do big data analytics reveal about neighbourhood organisational vitality? *Urban Studies* 58, 15 (2021), 3118–3139.
- [27] Scott Workman, Richard Souvenir, and Nathan Jacobs. 2015. Wide-Area Image Geolocalization with Aerial Reference Imagery. In *2015 IEEE International Conference on Computer Vision (ICCV)*. 3961–3969.
- [28] Abraham Noah Wu and Filip Biljecki. 2022. GANmapper: Geographical data translation. *International Journal of Geographical Information Science* 36, 7 (2022), 1394–1422.
- [29] Michael A. Wulder and Nicholas C. Coops. 2014. Satellites: Make Earth Observations Open Access. *Nature* 513, 7516 (2014), 30–31.
- [30] Zimin Xia, Olaf Booi, Marco Manfredi, and Julian F. P. Kooij. 2022. Visual Cross-View Metric Localization with Dense Uncertainty Estimates. In *Computer Vision – ECCV 2022*, Shai Avidan, Gabriel Brostow, Moustapha Cissé, Giovanni Maria Farinella, and Tal Hassner (Eds.). Springer Nature Switzerland, Cham, 90–106.
- [31] Yifang Yin, Wenmiao Hu, An Tran, Hannes Kruppa, Roger Zimmermann, and See-Kiong Ng. 2022. A Context-enriched Satellite Imagery Dataset and an Approach for Parking Lot Detection. In *2022 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. 1391–1400.
- [32] Yifang Yin, Rajiv Ratn Shah, Guanfeng Wang, and Roger Zimmermann. 2018. Feature-Based Map Matching for Low-Sampling-Rate GPS Trajectories. *ACM Trans. Spatial Algorithms Syst.* 4, 2, Article 4 (Aug. 2018), 24 pages.
- [33] Syed Waqas Zamir, Aditya Arora, Akshita Gupta, Salman H. Khan, Guolei Sun, Fahad Shahbaz Khan, Fan Zhu, Ling Shao, Gui-Song Xia, and Xiang Bai. 2019. iSAID: A Large-scale Dataset for Instance Segmentation in Aerial Images. *CoRR* abs/1905.12886 (2019).
- [34] Menghua Zhai, Zachary Bessinger, Scott Workman, and Nathan Jacobs. 2017. Predicting Ground-Level Scene Layout from Aerial Imagery. In *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 4132–4140.
- [35] Lichen Zhou, Chuang Zhang, and Ming Wu. 2018. D-LinkNet: LinkNet with Pre-trained Encoder and Dilated Convolution for High Resolution Satellite Imagery Road Extraction. In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*. 192–1924.
- [36] Zongwei Zhou, Md Mahfuzur Rahman Siddiquee, Nima Tajbakhsh, and Jianming Liang. 2018. UNet++: A Nested U-Net Architecture for Medical Image Segmentation. In *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support*. Springer International Publishing, Cham, 3–11.
- [37] Sijie Zhu, Taojiannan Yang, and Chen Chen. 2021. Revisiting Street-to-Aerial View Image Geo-localization and Orientation Estimation. In *2021 IEEE Winter Conference on Applications of Computer Vision (WACV)*. 756–765.
- [38] Sijie Zhu, Taojiannan Yang, and Chen Chen. 2021. VIGOR: Cross-View Image Geo-localization beyond One-to-one Retrieval. In *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 5316–5325.